

Task 3: Protein Structure Prediction

Tool name: Using **SWISS-MODEL** to predict the 3D structure of **Human Lysozyme (P61626)**.

Introduction: Proteins are fundamental molecules of life, and their three-dimensional (3D) structures are directly linked to their biological functions. SWISS-MODEL is a fully automated protein structure homology-modelling server hosted on the ExPASy web platform. The purpose of this server is to make protein modelling accessible to all life science researchers worldwide. The **PDB format** is downloaded from the **RCSB PDB**, and Human Lysozyme (P61626) is visualized in PyMOL.

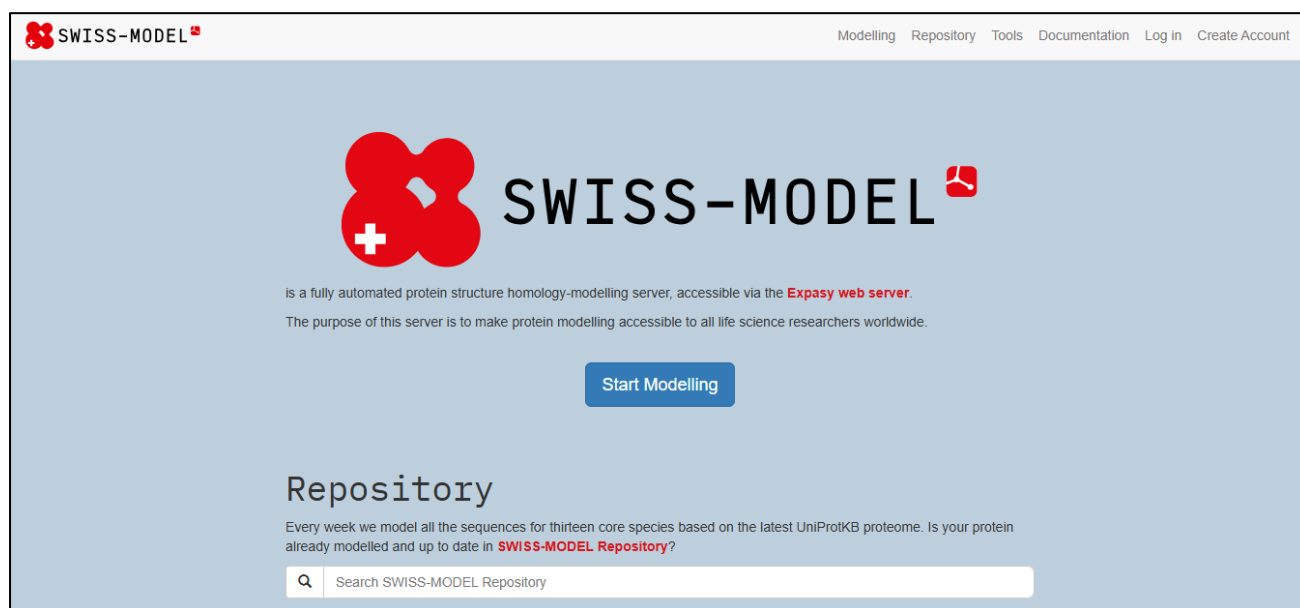


Fig 1: SWISS-MODEL homepage (<https://swissmodel.expasy.org/>).

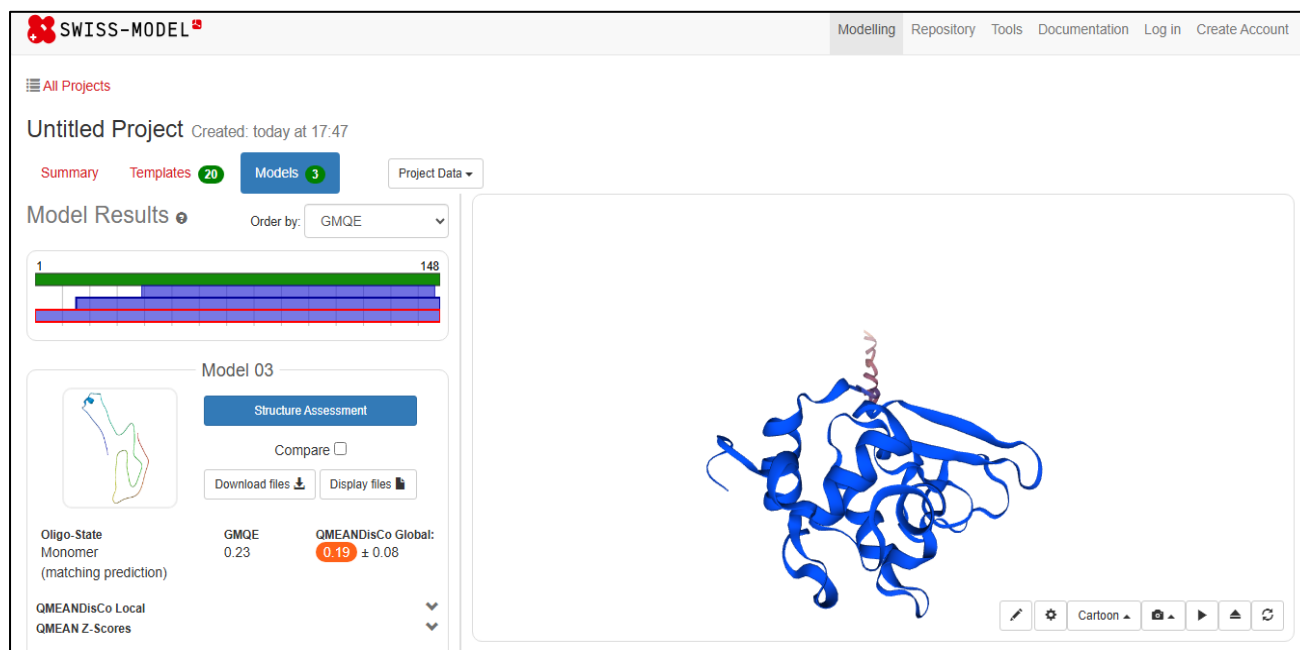


Fig 2: Predicted 3D structure of P61626 on SWISS-MODEL.

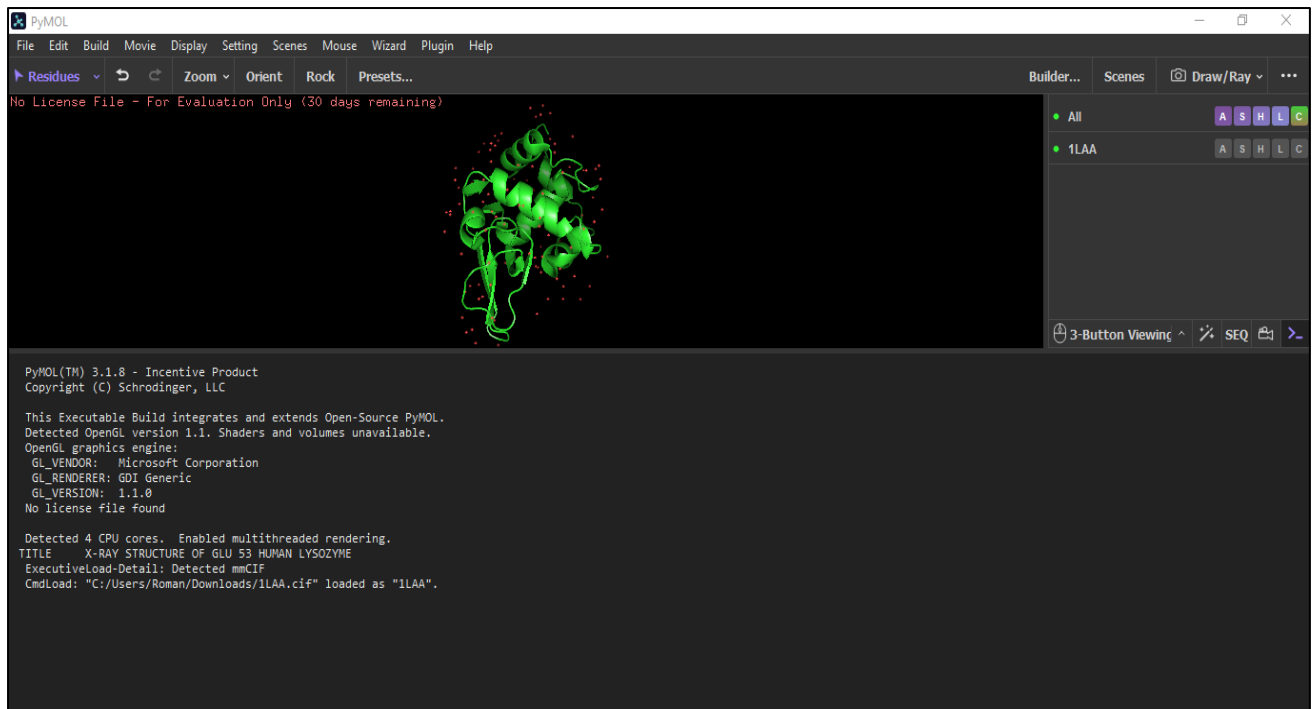


Fig 3: Visualization of P61626 3D structure in PyMOL.

Result

The three-dimensional structure of the **Human Lysozyme** protein is successfully predicted using the **SWISS-MODEL** server. The generated model offers valuable insights into the protein's folding patterns, structural stability, and potential functional sites, such as active domains or binding regions. Then, **PyMOL** is used for high-quality protein structure visualization of **P61626**.